MILITARY SPECIFICATION

FORCEPS, TISSUE, DEBAKEY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. Tweezer type tissue forceps covered by this specification are for use in cardiovascular surgery.

1.2 Classification. Forceps shall be of the following sizes as specified (see 6.2):

Size	Length
1	7-3/4 inches
2	9-1/2 inches

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

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STANDARDS

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards

E-18 Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 General requirements. The individual part requirements shall be as specified herein and in accordance with the applicable requirements of GG-I-526. In the event of any conflict between the requirements of GG-I-526 and those of this specification, the latter shall govern.

3.2 Materials. Materials shall be as specified herein.

3.2.1 Metal parts. The forceps shall be fabricated from Class 4, Type 410, 416 or 420 stainless steel as specified in GG-I-526, except that as an alternate the minimum carbon content of Type 420 material may be 0.14 percent.

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3.3 Design and construction. The instrument shall be a tweezer type, straight tissue forceps with rounded tips of the DeBakey design. It shall have 1 by 2 teeth and shall be designed to provide a firm, non-slip, atraumatic, triangular grasp.

3.3.1 Teeth. The profile of the teeth shall be well defined. With fully closed jaws there shall be a free slot of 0.003 inch at the outer edges of the tooth profile. Figure 2 delineates tooth profile dimensions and shall be used for tooling purposes only.

3.3.2 Physical dimensions. The physical configuration and dimensions of the instrument shall be in accordance with Figure 1.

3.3.3 Rockwell hardness. After proper stress relieving, the Rockwell hardness of finished instruments fabricated from Type 410 or Type 416 material shall be C36 to C41 and C42 to C47 for Type 420 material, when tested as specified in 4.5.1.

3.3.4 Corrosion resistance. The instrument shall be corrosion resistant conforming with the requirements of GG-I-526 when tested as specified in 4.5.2.1 and 4.5.2.2.

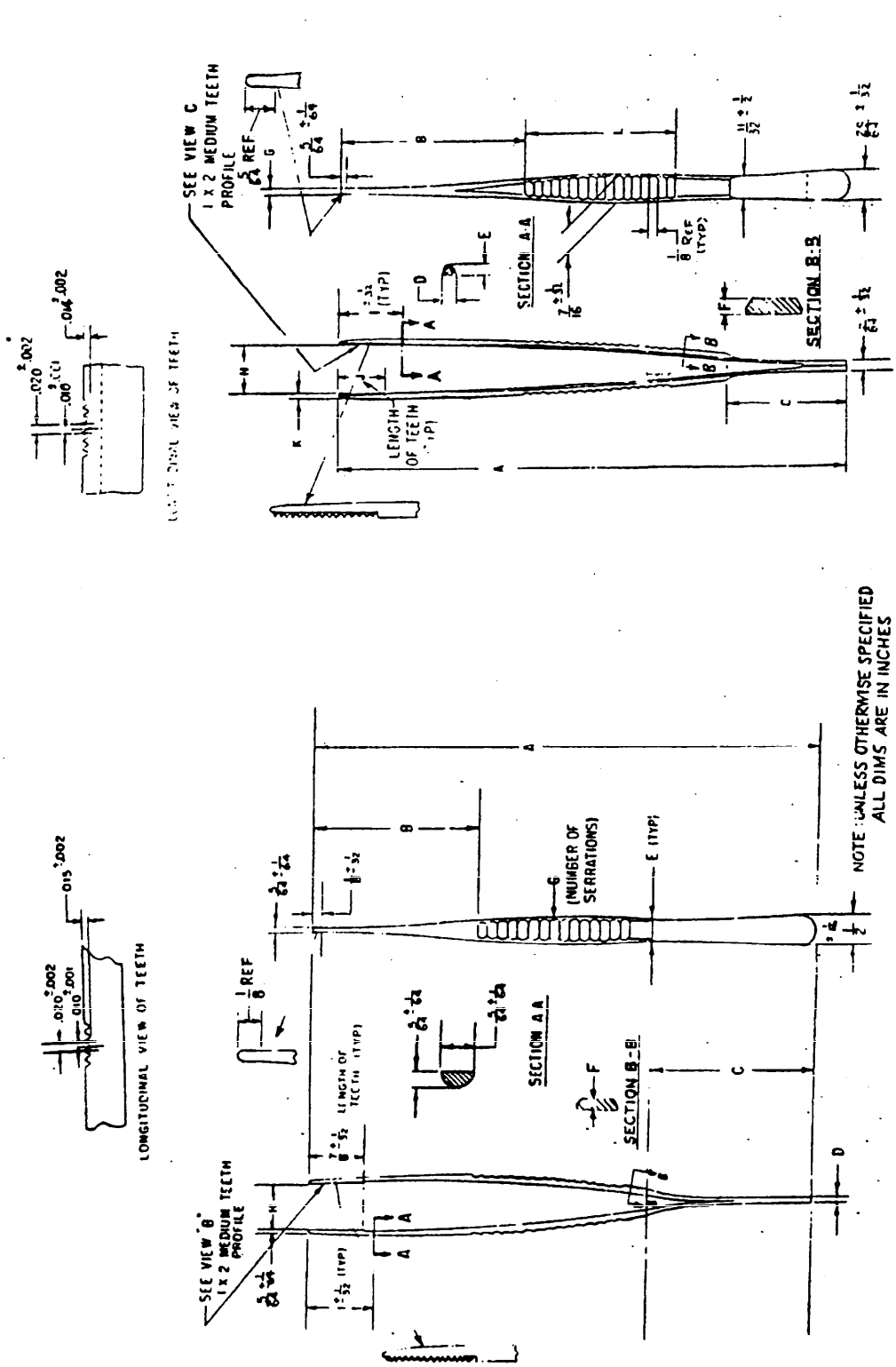
3.4 Finish. The finished instrument shall have a satin finish as specified in GG-I-526 and shall be non-reflective.

3.4.1 Passivation. As a final finish operation the complete instrument shall be passivated.

3.5 Identification marking. Identification marking of finished instruments shall be in accordance with GG-I-526. Instruments other than those manufactured in the United States shall also be marked "stainless".

3.6 Workmanship. Workmanship shall be first class throughout. The instrument shall be processed in such a manner as to be uniform in quality and shall be free from surface roughness, burrs, pits, scale, corrosion or any other defects that will affect life, performance, serviceability or appearance.

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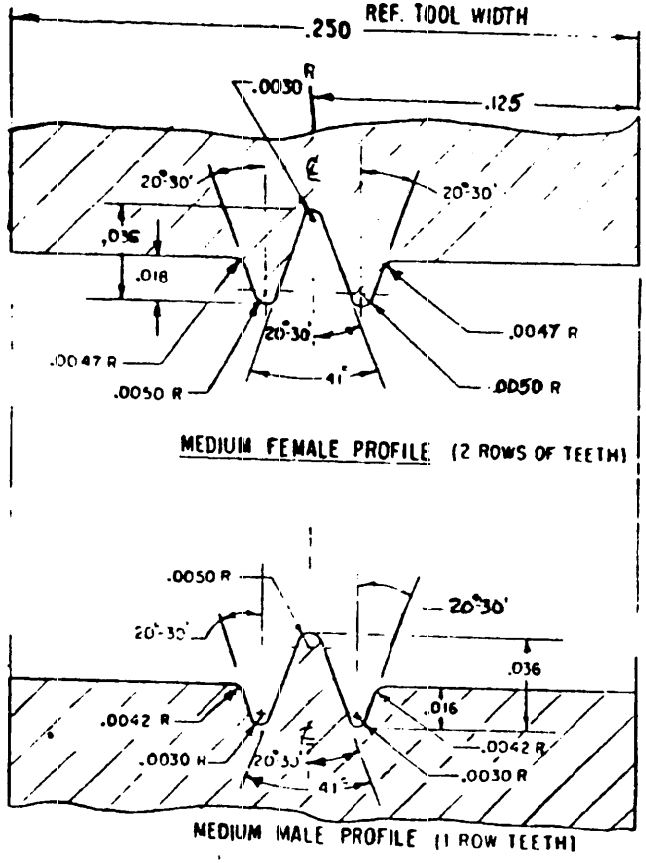
NSN	A ±1/64	B ±1/8	C ±1/8	D ±.004	E ±.004	F ±1/64	G ±.003	H ±1/64	J ±3/64	K ±.004	L ±1/8
6515-00-690-3208	7 7/8	2 3/4	2	0.114	0.079	3/32	0.087	51/64	23/32	0.059	2 3/8
6515-00-690-3209	9 1/2	3 7/16	2 3/16	0.118	0.087	7/64	0.094	61/64	51/64	0.067	2 3/4

ALTERNATE DESIGN

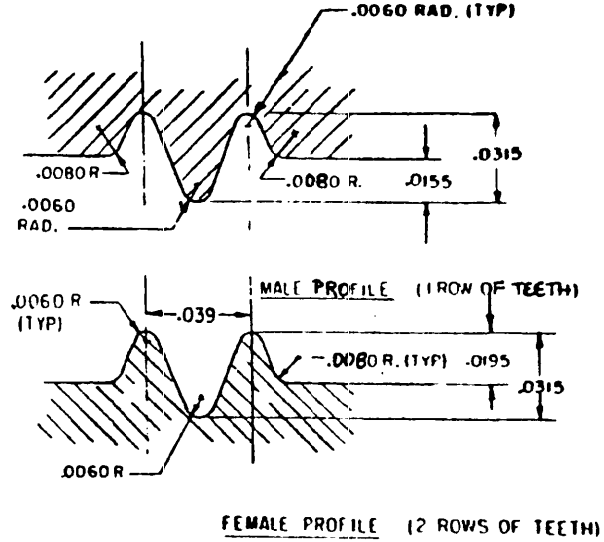
NSN	A ±1/64	B ±1/8	C ±1/8	D ±.004	E ±.004	F ±1/64	G ±.003	H ±1/64	J ±3/64	K ±.004	L ±1/8
6515-00-690-3208	7 7/8	2 3/4	2	0.114	0.079	3/32	0.087	51/64	23/32	0.059	2 3/8
6515-00-690-3209	9 1/2	3 7/16	2 3/16	0.118	0.087	7/64	0.094	61/64	51/64	0.067	2 3/4

FIGURE 1. Forceps, tissue, Debakey

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VIEW B



VIEW C

1 x 2 TEETH PROFILE

ALTERNATE DESIGN

NOTE:

1 - TEETH PROFILE DIMENSIONS SHOWN ARE FOR TOOLING PURPOSES ONLY
2 - WHEN JAWS ARE CLOSED IN THE NORMAL POSITION, TEETH SHALL NOT TOUCH; .002 TO .003 INCH SPACE SHALL EXIST.
3 - UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES

FIGURE 2. Dimensions, teeth, profile, tool

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the corrosion-resistant metals may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 Inspection lot. The inspection lot shall be determined in accordance with MIL-STD-105.

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4.2.2 Examination of preparation for delivery. An examination shall be made to determine that packaging and packing requirements comply with Section 5 of this specification. The inspection lot size shall be expressed in units of one fully prepared shipping container, containing complete instruments fully prepared for delivery and made essentially of the same materials and components. The sample unit shall be one fully prepared shipping container, containing instruments fully prepared for delivery, with the exception that it need not be sealed. The inspection shall be Inspection Level S-2 and the Acceptable Quality Level (AQL) shall be 4.0 percent defective in accordance with MIL-STD-105.

4.2.3 Sampling for visual and dimensional examination. A random sample of instruments shall be taken from each lot in accordance with MIL-STD-105. The Inspection level and AQL shall be as indicated in table I. The unit of product shall be one forceps.

TABLE I. Sampling for examination.

Type of defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-4	1.0
Minor	II	2.5

4.2.4 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Unless otherwise specified, the unit of product for testing purposes shall be one forceps.

TABLE II. Sampling criteria for tests.

Component Part	Characteristic	Requirement	Test Method	Inspection level	AQL
Forceps	Rockwell Hardness	3.3.3	4.5.1	S-2	1.5
	Corrosion resistance	3.3.4	4.5.2.1 and 4.5.2.2	S-2	1.5

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4.2.5 Raw materials. A sufficient amount from each lot of each metal used in the construction of the instrument shall be selected for chemical analysis.

4.3 Examination. The instrument shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with MIL-STD-105 and the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table III.

TABLE III. Classification of defects.

Categories	Defects
Critical	None defined.
Major A	
101	Style and design not as specified.
102	Instrument not constructed as specified.
103	Instrument not free from sharp edges, rough areas, pits, burrs, cracks, grind marks and overlaps.
104	Serrations not free of foreign matter or well defined.
105	Teeth profile not well defined and not of uniform depth.
106	Teeth do no mesh properly when forceps are closed.
107	Finish not as specified.
Major B	
151	Dimensions not as specified.
Minor	
201	Instrument not permanently and legibly marked with either the name or trademark of the contractor.

4.4 Examination of preparation for delivery. Each of the fully prepared shipping containers, containing instruments, selected as a sample unit from the lot shall be visually examined to determine that the packaging, packing and marking conform to this specification. The classification of defects, table IV, shall be used to classify the defects found.

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4.4.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

TABLE IV. Examination of defects of preparation for delivery.

Examine	Defect
Markings	Omitted, incorrect, illegible, or improper size, location, sequence or method of application.
Materials	Any component missing; any component damaged, affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Contents Exterior & Interior	Number of intermediate packages is more or less than required; gross/net weight exceeds requirements.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Hardness. Rockwell hardness shall be determined in accordance with ASTM test method E18. A minimum of 5 hardness readings shall be made on test specimens representative of the material used in the fabrication of the instrument or on the completed end item.

4.5.2 Corrosion tests.

4.5.2.1 Boil test. The instrument shall be subjected to the boil test as described in GG-I-526.

4.5.2.2 Copper sulfate test. The instrument shall be subjected to the copper sulfate test as described in GG-I-526.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each forceps shall be thoroughly cleaned and dried in accordance with the applicable method in MIL-P-116. Each forceps shall be wrapped in neutral tissue or other non-corrosive material and packaged in accordance with MIL-P-116, method IC-2. A preservative shall not be applied to the forceps.

5.1.1.2 Intermediate package. Twelve unit packages, containing forceps of one size only, shall be packaged in a snug fitting box constructed in accordance with PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure of the box shall be adequate to prevent accidental opening.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twenty-four intermediate packages (288 units) containing forceps of one size only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601 overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

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5.2.2 Level B. Twenty four intermediate packages (288 units) containing forceps of one size only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tissue forceps are intended for use in cardiovascular surgical procedures.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Applicable National Stock Number.

6.3 This specification is intended to cover the following items appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-00-690-3208	FORCEPS, TISSUE, Debakey, Straight, Round Tips, 7-3/4 inch
6515-00-690-3209	FORCEPS, TISSUE, Debakey, Straight, Round Tips, 9-1/2 inch

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6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1973

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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